

HALIFAX WINDSOR PARK, NS, Canada

WMO: 713270

Lat: 44.6570N

Lon: 63.6080W

Elev: 167

StdP: 14.61

Time Zone: -4.00 (NAA)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	8.8	-5.2	4.2	8.8	-2.2	4.9	12.6	22.6	32.1	20.4	30.4	8.2	280	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.6	82.3	67.4	79.3	65.9	76.7	64.7	71.0	77.2	69.3	75.1	67.8	73.0	7.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	107.7	73.4	67.5	101.6	71.7	65.9	96.0	70.1	35.0	77.1	33.5	75.1	32.3	73.0	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.6	14.4	DB	1.2	87.4	5.4	3.4	-2.7	89.8	-5.8	91.8	-8.9	93.7	-12.8	96.2	(4)
(5)			WB	-0.2	73.5	4.8	1.4	-3.7	74.6	-6.5	75.4	-9.1	76.2	-12.6	77.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	26.2	26.6	32.1	41.9	50.9	59.4	68.0	68.1	61.9	51.9	41.7	32.0		
	DBStd	16.41	9.66	8.63	7.90	6.18	5.67	5.73	4.46	3.87	5.69	6.63	7.75	9.08		
	HDD50	3141	738	654	558	253	55	4	0	0	0	56	265	558		
	HDD65	6903	1203	1074	1021	694	440	183	21	15	123	407	699	1023		
	CDD50	1989	0	0	3	9	82	286	557	561	358	116	16	1		
	CDD65	273	0	0	1	0	1	15	113	110	31	2	0	0		
	CDH74	1264	0	0	5	1	31	126	559	429	109	4	0	0		
	CDH80	222	0	0	1	0	6	26	107	61	21	0	0	0		
	WSAvg	6.9	8.0	8.0	8.2	7.7	6.7	6.1	5.5	5.4	5.8	6.6	7.1	7.6		
	PrecAvg	55.7	5.2	4.5	4.8	4.3	4.4	4.1	4.0	3.8	3.8	5.0	5.5	5.9		
(15) Precipitation	PrecMax	70.7	10.7	9.0	9.6	8.2	8.4	13.7	9.0	12.6	8.9	10.1	10.2	10.0		
	PrecMin	38.7	1.7	1.3	1.8	0.7	1.0	1.3	0.5	1.0	1.1	1.3	1.5	2.6		
	PrecStd	7.8	1.9	1.9	1.8	1.7	1.8	2.2	1.9	2.4	1.8	2.0	2.2	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	51.4	60.0	65.8	79.3	83.8	88.1	85.7	81.9	72.7	62.8	56.6		
		MCWB	52.5	49.0	50.5	51.0	60.7	66.1	70.3	70.2	69.1	61.3	60.4	55.4		
	2%	DB	48.4	45.8	50.3	59.6	69.4	77.1	82.7	81.1	76.9	67.7	59.6	52.3		
		MCWB	46.9	43.7	42.9	48.7	55.8	62.6	67.7	67.4	66.3	60.9	56.8	51.0		
	5%	DB	44.3	42.2	46.4	55.7	64.6	73.6	79.5	78.5	73.3	64.3	57.0	49.6		
		MCWB	42.5	39.9	41.0	47.4	53.6	60.7	66.1	65.8	64.5	59.0	54.9	48.3		
	10%	DB	39.9	38.5	42.8	52.0	60.6	70.0	76.9	76.0	70.5	61.8	54.0	46.0		
		MCWB	37.8	36.3	38.1	45.7	50.8	59.1	65.2	64.6	63.4	57.7	51.3	44.3		
	0.4%	WB	52.5	49.6	52.0	56.2	63.5	68.9	73.1	72.8	71.8	67.4	61.1	55.7		
		MCDB	53.4	50.2	57.9	60.9	73.7	80.1	80.7	79.4	77.7	69.6	62.5	56.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.2	44.0	45.6	52.8	59.2	65.6	71.0	71.0	69.3	63.3	57.9	51.3		
		MCDB	48.0	45.2	47.8	56.0	66.0	72.3	78.4	76.3	72.7	65.4	59.1	52.0		
	5%	WB	42.6	39.7	42.1	49.5	56.0	63.2	69.1	69.3	67.4	60.9	55.2	48.2		
		MCDB	43.7	41.1	45.4	52.5	60.9	70.0	76.0	74.4	70.9	63.0	56.6	49.3		
	10%	WB	38.0	36.4	38.7	46.6	53.2	61.0	67.5	67.7	65.5	58.3	51.8	44.6		
		MCDB	39.6	38.0	42.0	50.6	57.8	67.2	73.4	72.8	69.0	61.0	53.5	45.9		
	5% DB	MDBR	13.4	13.2	13.6	14.3	15.3	15.4	14.9	14.6	14.3	12.6	12.1	12.5		
		MCDBR	18.3	16.7	18.1	20.9	22.1	22.1	19.5	18.1	16.7	14.8	15.2	16.8		
		MCWBR	18.5	16.1	13.5	12.8	12.1	10.5	8.2	7.9	9.3	9.9	14.6	17.8		
		MCWBR	17.8	17.1	16.9	17.0	19.1	19.3	16.1	14.4	13.6	12.6	14.7	16.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	18.7	17.0	15.2	13.8	12.8	11.0	8.6	8.2	9.6	11.6	15.6	17.9		
	Clear-Sky Solar Irradiance	taub	0.302	0.295	0.323	0.348	0.368	0.397	0.413	0.394	0.353	0.331	0.314	0.295		
		taud	2.469	2.523	2.476	2.449	2.436	2.380	2.355	2.405	2.496	2.556	2.554	2.537		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	260	287	291	290	286	277	271	272	277	269	255	253		
		Edh at Noon	22	25	30	34	35	37	38	35	30	24	21	19		
(44) All-Sky Solar Radiation	RadStd		455	738	1116	1365	1569	1707	1701	1585	1288	835	479	358		
			35	55	76	113	200	159	124	110	86	87	51	27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	+0.65	+0.90	N/A	N/A	+90	+25

Nomenclature: See separate page